

Adjudicating competing theories of semantic representation with 7T-fMRI and multivariate decoding

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Abstract: Theories of semantic representation in the brain are numerous and contradictory and are supported by an equally contradictory literature of multivariate neuroimaging studies. Equipped with two pioneering tools – (1) a 7T-fMRI acquisition sequence capable of capturing signal in crucial ventral temporal regions, and (2) four innovative decoding methods that are designed to probe specific hypotheses about the nature of the semantic code – we resolve this confusion. Evidence best supports the hypothesis that the ventral anterior temporal lobe represents complex, multidimensional semantic information via an interdependent, dynamic code similar to that observed in neural networks. Similar representations – highly variable across individuals – were discovered in posterior temporal and occipitotemporal cortex. In summary, this work adjudicates competing theories of semantic cognition, thereby reconciling the discrepancy in results between multivariate imaging and other sources of evidence – neuropsychology, noninvasive brain stimulation, intracranial electrophysiology, and neural network modelling.

Keywords: semantic representation, 7T-fMRI, decoding, multivariate pattern analysis

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1. Introduction

Semantic cognition is our ability to store general knowledge and to use this knowledge to interpret the meaning of language, objects, and events. Influential theories of how the brain might encode semantic information are numerous and contradictory^{1–9} and are supported by an equally contradictory literature of multivariate neuroimaging studies^{10–18}. This confusion arises for at least two reasons – (1) magnetic field inhomogeneity^{19–26} that compromises signal quality in crucial ventral temporal regions and (2) analytic assumptions, embedded in multivariate methods, that restrict the kinds of hypotheses that each method can test²⁷. This study surmounted these obstacles by (1) utilising a novel 7 Tesla functional magnetic resonance imaging (7T-fMRI) acquisition protocol capable of recovering signal in regions prone to imaging artefacts²⁶ and (2) employing four innovative decoding methods^{11,28–31} that are designed to directly adjudicate different hypotheses about the nature of the semantic code. Previous theoretical work described what semantic representations might possibly be like²⁷ – here, for the first time, we resolve those outstanding questions empirically.

Despite their many differences, multiple theories of semantic cognition agree that the ventral anterior temporal lobes (vATLs) are important (cf.^{7,8}). This claim is supported by multiple sources of evidence – in semantic dementia, atrophy of the vATL causes multimodal semantic impairment but spares other cognitive abilities^{32–35} and a similar, though milder and transient, effect is observed following transcranial magnetic stimulation (TMS) to the vATL^{23,36–39}. However, theorists disagree on the vATL’s precise role. Damasio and colleagues³ propose that modality-specific semantic information (an object’s visual appearance, name, associated motor action, etc.) is encoded in separate modality-specific regions; convergence zones – one for each concept – trigger and synchronise activity within these areas during representation. The convergence zones are organised, by category, into convergence regions, with a convergence region for unique entities (such as individuals) in the temporal pole and a convergence region for animals just behind⁴. By contrast, Binder and colleagues² propose that the vATL functions, not simply as a pointer towards modality-specific cortex, but as one of multiple semantic “hubs” that computes complex semantic representations from independent features (e.g. *has eyes*, *has gills*) represented elsewhere^{1,2,40}. More similar concepts share more features, accounting for graded semantic

similarity (e.g. a robin and a wren are more similar to each other than to an ostrich). A third theory, the hub-and-spoke model^{6,9,41}, is motivated by the finding that neuro-computational models with a single hub can represent transmodal semantic structure (whereas other architectures fail^{9,42,43}) and proposes that the vATL is the brain's *only* semantic hub. Like the deep layers of a computational model, activity within the vATL does not correspond to interpretable features – rather, after initial dynamic and nonlinear change, activity patterns adopt steady states that are gradedly more similar for more similar concepts.⁴⁴

Notwithstanding its theoretical importance, neuroimaging studies of semantic tasks rarely highlight the vATL. One possible reason is the proximity of the vATL to the air-filled sinuses – this causes magnetic field inhomogeneity, and consequently signal dropout and distortions. Univariate fMRI studies that use distortion correction, such as spin-echo or multi-echo acquisition, do detect semantic effects in the vATL^{19–26}. A second possible reason concerns analysis. Multivariate methods – which analyse patterns of activity over multiple neural units (voxels, sensors, or electrodes) to characterise what information those patterns represent – each encapsulate hypotheses that constrain the kind of neural code that they can detect and these assumptions can lead to drastically different results²⁷. For example, when decoding semantic information from voltages recorded from electrocorticography (ECoG) surface grid electrodes, representational similarity analysis⁴⁵ revealed a simple binary animacy (living/non-living) distinction in the ATL⁴⁶ whereas methods based on regularised regression revealed a complex, multidimensional semantic structure²⁸.

Each contemporary theory of semantic cognition also implicates regions outside the vATL. Several theories agree that modality-specific regions are important because they encode modality-specific semantic information about a concept^{2,3,6–8}. For example, neuroimaging studies highlight the inferior parietal cortex when processing tools⁴⁷; lesions to inferior parietal regions can cause category-specific semantic impairments for tools⁴⁸; and similar but transient impairments can be achieved with TMS³⁹. Additionally, neuroimaging studies report activity in posterior medial temporal cortex when processing tools and in posterior lateral temporal cortex when processing animals^{7,49}. However, theorists interpret this evidence differently – some assert that this indicates representation of interpretable features belonging to animals and tools; others note that the same results emerge in computational models with uninterpretable activations due to differential connectivity patterns of posterior temporal subregions⁵⁰. Theories that propose the existence of multiple

hubs assign a central role to the angular gyrus, which is highlighted by imaging studies of semantic tasks^{1,51} and which, when damaged, produces a pattern of impairment that appears to be mostly semantic⁵² (cf.^{53,54}). Finally, some theories even propose that semantic information is radically dispersed across most or all of the cortex^{5,14,15}. Each of these opposing theories is supported to some extent by multivariate neuroimaging studies^{10,12–15,28,44}. Although regions outside the ventral anterior temporal and orbitofrontal cortices are not affected by magnetic field inhomogeneity, methodological choices (which incorporate theoretical assumptions) may contribute to this variability.

In summary, contemporary theories of semantic cognition make contradictory claims. These claims concern not only *where* semantic information is represented, but also *how* it is represented – for example, Binder and colleagues² argue that semantic representations are constructed through linear combination of interpretable features, whereas the hub-and-spoke model^{6,9,41} proposes that semantic features are uninterpretable and interdependent. Most multivariate neuroimaging studies test the predictions of a single theory; this work is distinctive because we directly tested the predictions of competing theories against one another. Utilising a clear theoretical framework²⁷, we tested whether semantic representations are:

1. *Category* representations (one discrete, independent unit per concept), *feature-based* representations (multiple independently-interpretable features), or *vector-space* representations (patterns across units that are not independently interpretable)
2. *Homogeneous* (units all adopt the same activation state – i.e., all increase in activity or all decrease – when representing the same concept) or *heterogeneous* (units adopt different activation states for the same concept)
3. *Consistent* (units encode semantic information in the same way across individuals) or *inconsistent* (units exhibit different semantic codes across individuals)
4. *Contiguous* (units are spatially adjacent) or *dispersed* (units are spatially distributed across the cortex) – representations can be contiguous or dispersed within or across individuals

We also assessed the compatibility of our results with the hub-and-spoke model's claim that representations in the vATL are *dynamic*, meaning that the information carried by a

particular neural unit changes over time – for example, a unit may signify animacy with increased activity at one timepoint and with decreased activity at the next⁴⁴.

We overcame two obstacles that have hindered previous attempts to characterise semantic representations. First, to address magnetic field inhomogeneity affecting ventral temporal regions, we employed a novel multi-echo, multiband acquisition sequence optimised for recovering signal in the vATL without sacrificing signal quality elsewhere²⁶. Second, we leverage the inseparability of multivariate analysis and theory to our advantage – we employ multivariate methods that make opposing theoretical assumptions and then contrast the results obtained with each method. First, we decoded animacy using binary logistic regression classifiers. These were used in combination with LASSO (L1) regularisation⁵⁵, which assumes that the neural code is sparse (only a few of the features under consideration encode the target structure), and with sparse-overlapping-sets LASSO (SOSLASSO) regularisation^{11,30,31}, which additionally assumes that important features are loosely spatially clustered within and across participants. Second, we used Representational Similarity Learning (RSL)^{28,29} to decode graded, multidimensional semantic similarity structure. In RSL, a similarity structure is modelled (for example, using feature norms^{27,56}), then decomposed into multiple independent dimensions. Linear regression models then learn to predict the coordinates of each stimulus on each dimension; significant correlations between true and predicted coordinates indicate that the modelled similarity structure is a good representation of the brain's. With RSL we used LASSO regularisation and group-ordered-weighted LASSO (grOWL) regularisation^{28,29}, which assumes that the neural code is sparse, that important features are correlated in their activity, and that features are likely to encode multiple target dimensions at once (an interdependent, vector-space representation).

We applied regularised logistic regression and RSL within a vATL region of interest (ROI) and at the whole-brain level with the aim of adjudicating contemporary theories about the nature of semantic representation.

2. Results

2.1. Characterising semantic representations in the ventral anterior temporal lobe (vATL)

27 healthy participants completed the study successfully. None were excluded because of excessive head motion, as defined by two metrics – framewise displacement⁵⁷ and DVARS (which indexes the rate of change of BOLD signal across the entire brain between each pair of successive frames of data)⁵⁸. The participants named 100 line drawings of animals and inanimate objects that were used previously for decoding^{28,44} and that were matched for age of acquisition, concreteness, word frequency and familiarity. The participants named the stimuli with high accuracy (mean = 96.85%, standard deviation 2.71%). Three participants had at least one run with accuracy less than two standard deviations below the group mean. However, since none of these participants had accuracy less than 88% on any run, we decided to include them.

Multiple theories propose that the vATL plays an important role in semantic cognition^{2,3,6,9,41}, although its precise role is debated – a convergence region³ or a hub, combining independent^{2,7,8} or interdependent features^{6,9,41}. Therefore, we aimed first to decode semantic information from the vATL and next to characterise the nature of semantic representations there. We constructed bilateral ventral temporal ROIs based on ECoG electrode coverage in a previous decoding study⁵⁹. We also created “half-ROIs” by bisecting the ROIs between their anteriormost and posteriormost points. This was primarily to facilitate tests for dynamic coding, which was previously observed more distinctly in anterior than posterior temporal regions in ECoG data⁴⁴, but for completeness we ran all decoding analyses on all ROIs and half-ROIs. The ROIs were backprojected into each participant’s native space and were masked with a participant-specific grey matter mask.

All models were trained on contrast beta values (one per stimulus, averaged over repeated presentations of the same stimulus) within each ROI and were assessed using ten-fold cross-validation.

2.1.1. Is semantic structure present in the vATL?

2.1.1.1. Animacy

First, we attempted to decode animacy from the vATL using logistic regression. Figure 1A shows the results. All classifiers trained on all ROIs performed better than chance.

Classification accuracy was very high (hold-out accuracy $\cong 0.95$) for decoders trained on the whole ROI or on the posterior half. This result confirmed that animacy was represented within our ROI.

2.1.1.2. Multidimensional semantic structure

Second, we attempted to decode multidimensional semantic structure from the vATL using RSL. We modelled a target semantic similarity structure using feature norms⁵⁶ and used singular value decomposition to decompose the structure into three dimensions that explained 89.5% of the variance. Figure 2 shows the coordinates of all 100 stimuli on each dimension. We then trained RSL models to predict the coordinates of each stimulus on all dimensions.

Figure 1B shows the mean hold-out correlation between predicted and true coordinates on each target semantic dimension. All models showed very good decoding of dimension 1 of the target semantic space in all ROIs. In the right hemisphere, all models produced significant correlations on at least one other dimension that, though small, were statistically significant as assessed via permutation testing (see Methods; dimension 3 for LASSO, and both dimension 2 and dimension 3 for grOWL). In the left hemisphere, dimensions 2 and 3 were not reliably decoded in any ROI. These results confirmed the presence of multidimensional semantic structure in the right ROI.

2.1.1.3. Graded semantic structure

As shown in Figure 2, animate stimuli usually take low values on dimension 1 and inanimate stimuli usually take high values. Therefore, a model that predicted the same low value for all animate stimuli and the same high value for all inanimate stimuli could generate predicted coordinates that correlate reliably with the target coordinates, yet the model would not be detecting graded similarity structure. However, significant correlations *within* animate and inanimate domains (i.e. when the model is evaluated only on the animate or only on the inanimate stimuli) arise only if semantic structure is truly graded.²⁸

Figure 3 shows correlations calculated for animate and inanimate stimuli separately. For animate stimuli, results depended on the regularisation penalty - whereas models with

LASSO exhibited reliable decoding of only one dimension, models with grOWL discovered within-domain structure for dimensions 1 and 2. For inanimate stimuli, although correlation coefficients were generally smaller, multiple dimensions were decoded. Whereas these patterns were somewhat variable for models regularised with LASSO, models with grOWL systematically showed significant decoding of dimensions 1 and 3 in both hemispheres. These results confirmed that graded semantic structure was represented within our ROI.

When the two domains were considered separately, multidimensional semantic structure was observed in the left temporal lobe for both, despite the fact that correlations on only one dimension were observed there when all stimuli were considered together (Figure 1). Note that dimension 2 distinguishes the animate stimuli, but not the inanimate stimuli, and the reverse is true for dimension 3 (Figure 2); significant correlations within-domain usually emerged on dimension 2 for animate stimuli and on dimension 3 for inanimate stimuli.

2.1.2. What are semantic representations in the vATL like?

2.1.2.1. Categories, features, or vector-spaces?

Damasio and colleagues³ propose that the vATL contains category-specific convergence regions for animals and unique entities. This theory would predict successful decoding of animacy from our ROI – it would be active for animate stimuli and not for inanimate stimuli. However, because convergence zones simply point to representations elsewhere – they do not themselves express semantic information – the vATL would not be predicted to express multiple semantic dimensions. Figures 1 and 3 show that our results are at odds with this prediction.

Of the theories that do predict multidimensional information representation in the vATL, some propose that its role is to combine independently-interpretable features² while others propose that it encodes a vector-space with interdependent, uninterpretable dimensions^{6,9,41}. These possibilities can be adjudicated by comparing RSL results with different regularisation penalties. grOWL assumes that features are likely to encode multiple target dimensions at once whereas LASSO does not make that assumption. Therefore,

better performance with grOWL than LASSO supports the notion of interdependent, vector-space representation.

Figure 1B shows that RSL models with grOWL regularisation produced significant correlations on all three dimensions, whereas models with LASSO regularisation produced significant correlations on only two. Figure 3 shows that, for animate stimuli, models with LASSO exhibited reliable decoding of only one dimension; models with grOWL could decode dimensions 1 and 2. Within the inanimate domain, models with grOWL systematically showed significant decoding of dimensions 1 and 3 whereas LASSO exhibited less consistent results. Although grOWL produced larger correlations than LASSO (permutation $p < 0.05$) in only a few cases, the reverse was never true. These results suggest that the semantic code within the vATL is an uninterpretable, vector-space code.

2.1.2.2. Contiguous or dispersed?

SOSLASSO assumes that features are loosely spatially clustered within and across participants, but LASSO does not make that assumption. Figure 1A shows that, in all ROIs apart from the right anterior half-ROI, SOSLASSO produced significantly higher classification accuracy than LASSO (permutation $p < 0.05$), suggesting that semantic representations within the vATL are loosely contiguous within and across individuals.

2.1.2.3. Static or dynamic?

The hub-and-spoke model^{6,9,41} claims that activity patterns within the vATL undergo dynamic and nonlinear change during stimulus processing – for example, a voxel may signify animacy with increased activity at one timepoint and with decreased activity at the next. A previous ECoG study, focused on decoding animacy, observed these patterns more distinctly in very anterior than in more posterior temporal regions⁴⁴. The temporal resolution of fMRI is far lower than that of ECoG and the resulting spatial blurring may therefore obscure a temporally intricate semantic code. We would therefore expect poorer decoding performance in the anterior than the posterior half of the ROI. Figure 2A shows that classification accuracy in the anterior half-ROI was significantly worse than accuracy in the

posterior half-ROI (permutation $p < 0.05$). This fMRI result does not *establish* that the semantic code is dynamic, but the results are *compatible* with this claim.

Similarly, Figure 1B shows worse decoding in the anterior half-ROI than the posterior half ($r \cong 0.4$ vs. $r \cong 0.8$). However, dynamic representation of multidimensional semantics has not yet been demonstrated with time-resolved imaging methods. We therefore chose not to formally test the differences between half-ROIs or to interpret differences as evidence for dynamic coding of multidimensional semantic structure.

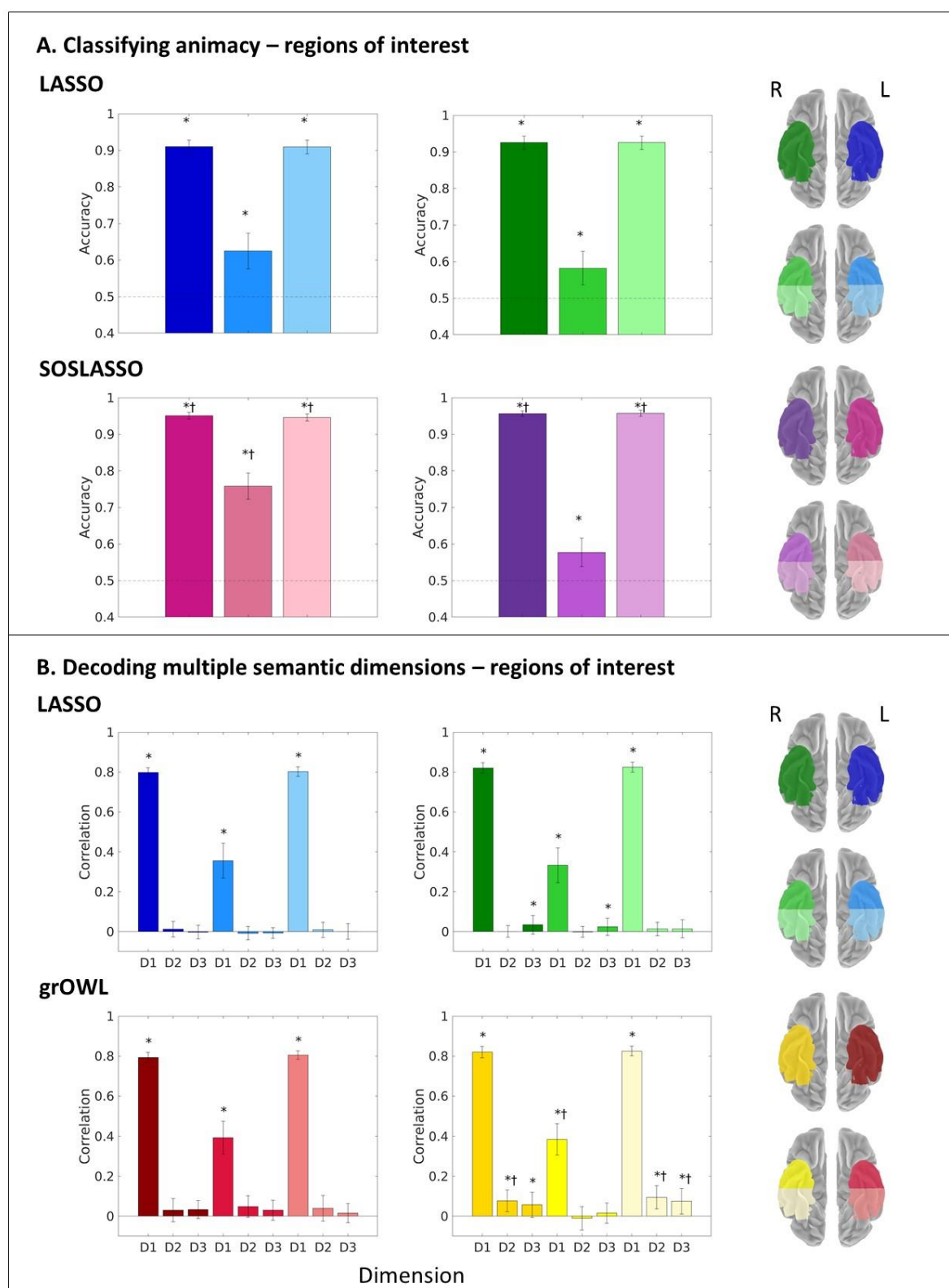


Figure 1: Decoding in regions of interest. Surface plots (right) indicate which bar corresponds to which ROI or half-ROI. (A) Mean and 95% confidence interval of the hold-out accuracy for regularised logistic regression classifiers trained on contrast beta values to discriminate animate from inanimate stimuli. Classifiers are trained using LASSO regularisation on the left ROI and half-ROIs (upper left, shades of blue), using LASSO regularisation on the right ROI and half-ROIs (upper right, shades of green), using SOSLASSO

regularisation on the left ROI and half-ROIs (lower left, shades of pink), or using SOSLASSO regularisation on the right ROI and half-ROIs (lower right, shades of purple). Stars (*) indicate a significant difference between classifier accuracy and chance (as determined by permutation test, probabilities adjusted to control the false-discovery rate at $\alpha = 0.05$). Daggers (†) indicate a significant difference between classifier accuracy using that regularisation penalty and accuracy in the same ROI using the other regularisation penalty (probabilities adjusted to control the false-discovery rate at $\alpha = 0.05$). (B) Mean and 95% confidence interval of the hold-out correlation for RSL models trained on contrast beta values to predict the coordinates of held-out stimuli on three target semantic dimensions (dimension 1, D1; dimension 2, D2; and dimension 3, D3). Models are trained using LASSO regularisation on the left ROI and half-ROIs (upper left, shades of blue), using LASSO regularisation on the right ROI and half-ROIs (upper right, shades of green), using grOWL regularisation on the left ROI and half-ROIs (lower left, shades of red), or using grOWL regularisation on the right ROI and half-ROIs (lower right, shades of yellow). Stars (*) indicate a significant correlation between predicted and target coordinates (as determined by permutation test, probabilities adjusted to control the false-discovery rate at $\alpha = 0.05$). Daggers (†) indicate a significant difference between correlation on that dimension using that regularisation penalty and correlation on the same dimension in the same ROI using the other regularisation penalty (probabilities adjusted to control the false-discovery rate at $\alpha = 0.05$).

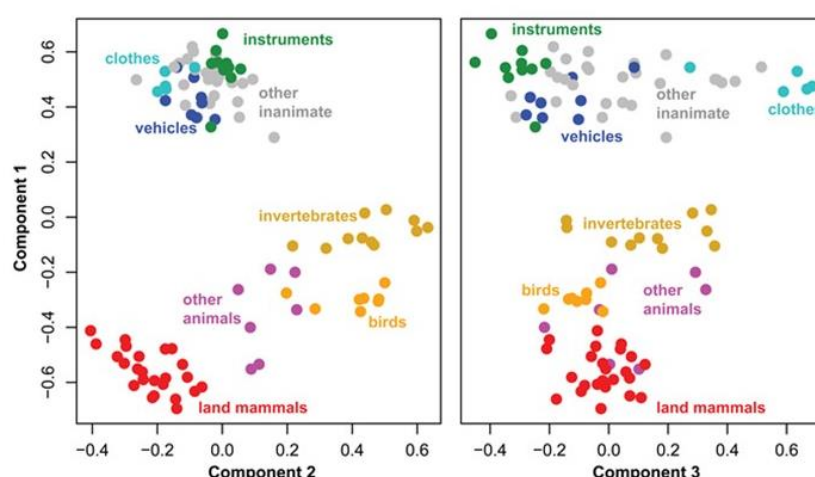


Figure 2: Coordinates of all stimuli on each target semantic dimension. Colours indicate category membership – land mammals (red), birds (orange), invertebrates (brown), other animals (magenta), instruments (green), clothes (light blue), vehicles (dark blue), and other inanimate objects (grey). Reprinted with permission from Cox and colleagues²⁸.

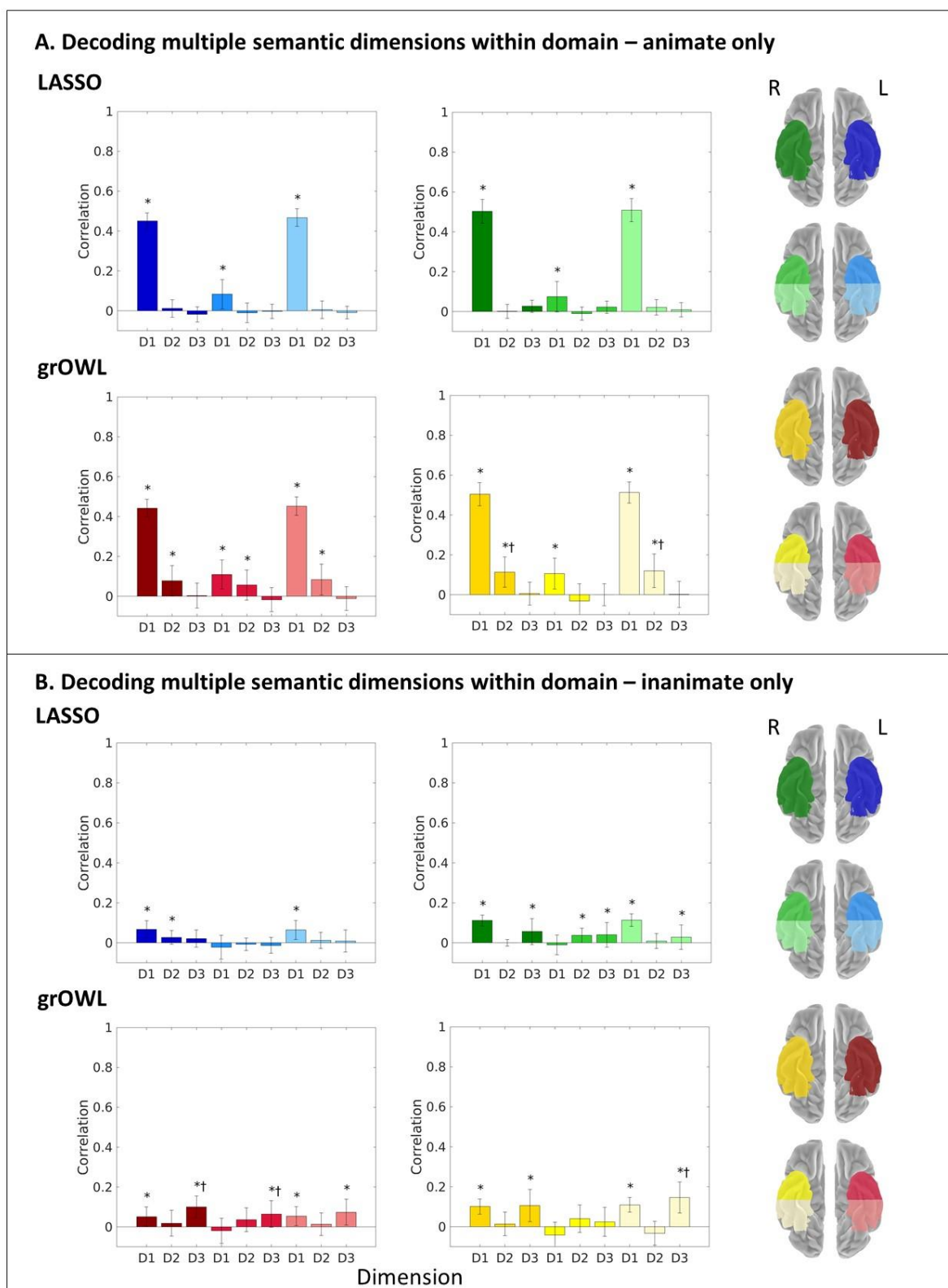


Figure 3: Hold-out correlations within-domain in regions of interest. Mean and 95% confidence interval of the hold-out correlation for RSL models trained on contrast beta values to predict the coordinates of held-out stimuli on three target semantic dimensions (dimension 1, D1; dimension 2, D2; and dimension 3, D3). Surface plots (right) indicate which bar corresponds to which ROI or half-ROI. Models are trained using LASSO regularisation on the left ROI and half-ROIs (upper left, shades of blue), using LASSO

regularisation on the right ROI and half-ROIs (upper right, shades of green), using grOWL regularisation on the left ROI and half-ROIs (upper left, shades of red), or using grOWL regularisation on the right ROI and half-ROIs (upper right, shades of yellow). Stars (*) indicate a significant correlation between predicted and target coordinates (as determined by permutation test, probabilities adjusted to control the false-discovery rate at $\alpha = 0.05$). Daggers (†) indicate a significant difference between correlation on that dimension using that regularisation penalty and correlation on the same dimension in the same ROI using the other regularisation penalty (probabilities adjusted to control the false-discovery rate at $\alpha = 0.05$). (A) Hold-out correlations for animate stimuli only; (B) Hold-out correlations for inanimate stimuli only.

2.2. Characterising semantic representations at the whole-brain level

All contemporary theories of semantic cognition implicate some regions outside the vATL – modality-specific regions^{2,3,6–8}, posterior temporal cortex^{7,49,50}, the inferior parietal lobe including the angular gyrus^{1,2}, or even the whole cortex^{5,14,15}. Therefore, we next aimed to investigate where else in the brain semantic information could be detected and to characterise the semantic representations there. For these analyses, all models were trained on contrast beta values (one per stimulus, averaged over repeated presentations of the same stimulus) from the whole brain. Models were assessed using ten-fold cross-validation.

2.2.1. Is semantic structure present at the whole-brain level?

2.2.1.1. Animacy

Figure 4A shows the results from decoding animacy with logistic regression with LASSO or SOSLASSO regularisation. As in the ROI analysis, classification accuracy was very high (hold-out accuracy $\cong 0.95$). This result confirmed the presence of information about animacy.

To assess which regions drove this result we visualised classifier coefficients on the cortical surface (see Methods) and compared these to univariate results (animate > inanimate and inanimate > animate; Figure 5A). Both univariate and multivariate analyses highlighted posterior temporal and occipitotemporal regions – the univariate map included some parietal regions that were absent from the multivariate maps and the multivariate maps encompassed the ventral posterior temporal lobe more comprehensively than the univariate map did.

2.2.1.2. Graded, multidimensional semantic structure

As shown in Figure 4B, decoding of dimension 1 was very successful ($r \cong 0.8$). However, evidence for *multidimensional* structure depended on the regularisation penalty - models with grOWL produced significant correlations on dimensions 2 and 3 whereas models with LASSO did not. The results from RSL with grOWL provided evidence for multidimensional semantic structure at the whole-brain level.

Figure 6 shows the hold-out correlations calculated for animate and inanimate stimuli separately. Although only RSL with grOWL revealed multidimensional structure when all stimuli were considered together, both RSL with LASSO and RSL with grOWL produced significant correlations on two dimensions within-domain. For animate stimuli, these were dimensions 1 and 2 (which separates out the animate stimuli) and, for inanimate stimuli, the dimensions were dimension 1 and 3 (which separates out the inanimate stimuli). These findings confirmed the presence of graded structure at the whole-brain level.

Figure 7 shows the coefficients from RSL models. As in the classification analyses, posterior temporal and occipitotemporal regions were selected. Other regions that have been proposed to represent semantic information, such as the angular gyrus^{1,2}, were absent from these maps. The vATL was also absent^{6,9,41}, despite the fact that our ROI analysis detected semantic structure there.

2.2.2. What are semantic representations outside the vATL like?

2.2.2.1. Categories, features, or vector-spaces?

Figure 4B shows that only RSL with grOWL was successful at decoding more than one dimension when all stimuli were considered together; Figure 6 shows that RSL with LASSO could successfully decode two dimensions within-domain, but was outperformed by RSL with grOWL when decoding dimension 2 for animate stimuli and dimension 3 for inanimate stimuli (permutation $p < 0.05$). Figure 7 shows that, when visualising coefficients on the cortical surface, models with grOWL yielded significant results for all three dimensions. However, for models with LASSO, coefficients for dimensions 2 and 3 did not survive

permutation thresholding. This clear advantage of grOWL over LASSO indicates that the representations in posterior temporal cortex are likely to be interdependent, vector-space representations¹⁵.

2.2.2.2. Homogeneous or heterogeneous?

The coefficient maps for the classification analysis (Figures 6B and 6C) feature lateral occipitotemporal regions in which coefficients are consistently negative across participants (meaning that, in most participants, a positive beta value is associated with an increased probability that the stimulus is animate) and medial occipitotemporal regions in which coefficients are consistently positive. The same pattern is seen in Figure 7B in the coefficients for dimension 1 in the RSL analysis with grOWL (this correspondence would be expected because dimension 1 distinguishes animate from inanimate stimuli; Figure 2). This demonstrates that these representations are heterogeneous – some units increase in activity when representing a concept; others decrease.

2.2.2.3. Consistent or inconsistent?

The coefficient maps for the classification analysis (Figures 6B and 6C) and for dimension 1 in the RSL analysis with grOWL regularisation (Figure 7B) feature regions in which coefficients are inconsistent in direction across participants. In the coefficient maps for dimensions 2 and 3, the regions of inconsistent coefficient directions were even larger.

2.2.2.4. Contiguous or dispersed?

In contrast to the ROI analysis, there was no difference in classification accuracy between SOSLASSO and LASSO (Figure 4A) and the same regions were highlighted with both methods (Figures 6B and 6C). This does not constitute evidence for contiguity within and across participants, although note also that classification performance may have been at ceiling and that this may account for the similarity between methods.

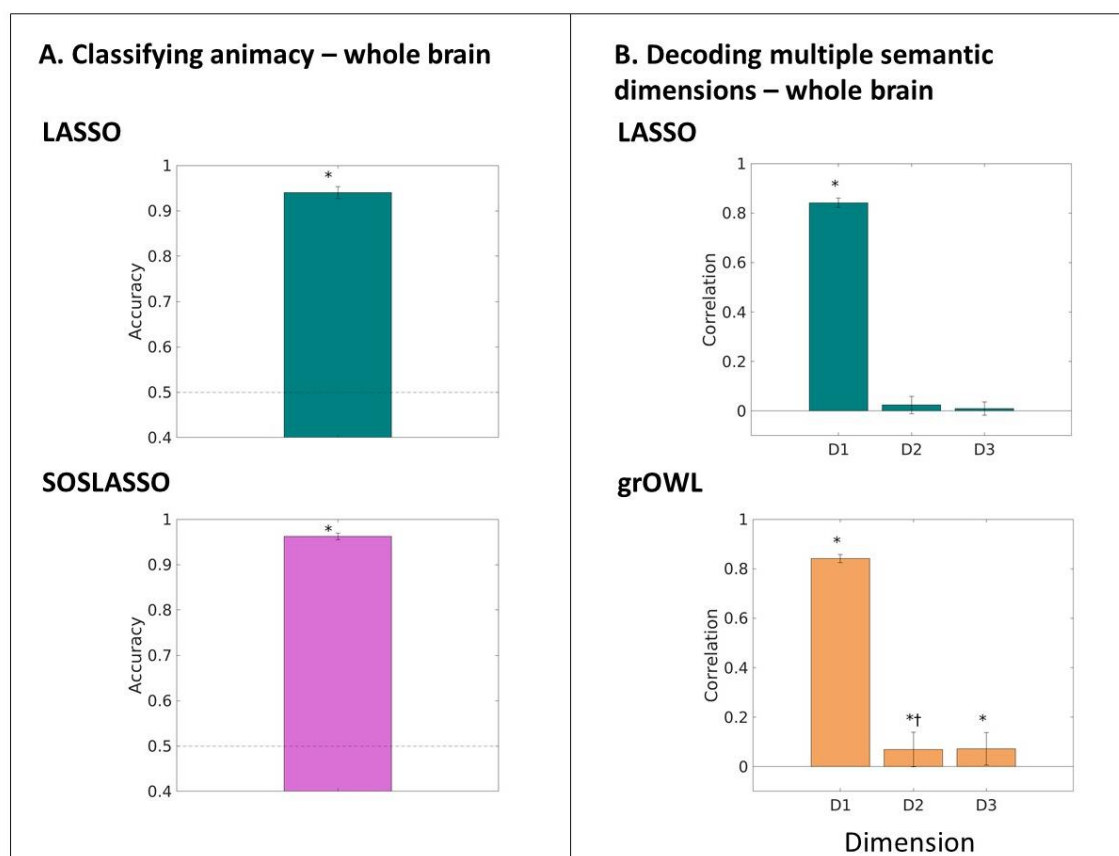


Figure 4: Decoding results at the whole-brain level. (A) Mean and 95% confidence interval of the hold-out accuracy for regularised logistic regression classifiers trained on contrast beta values to discriminate animate from inanimate stimuli. Classifiers are trained using LASSO regularisation (top, teal) or using SOSLASSO regularisation (bottom, magenta). Stars (*) indicate a significant difference between classifier accuracy and chance (as determined by permutation test, probabilities adjusted to control the false-discovery rate at $\alpha = 0.05$). (B) Mean and 95% confidence interval of the hold-out correlation for RSL models trained on contrast beta values to predict the coordinates of held-out stimuli on three target semantic dimensions (dimension 1, D1; dimension 2, D2; and dimension 3, D3). Models are trained using LASSO regularisation (top, teal) or using grOWL regularisation (bottom, orange). Stars (*) indicate a significant correlation between predicted and target coordinates (as determined by permutation test, probabilities adjusted to control the false-discovery rate at $\alpha = 0.05$). Daggers (†) indicate a significant difference between correlation on that dimension using that regularisation penalty and correlation on the same dimension using the other regularisation penalty (probabilities adjusted to control the false-discovery rate at $\alpha = 0.05$).

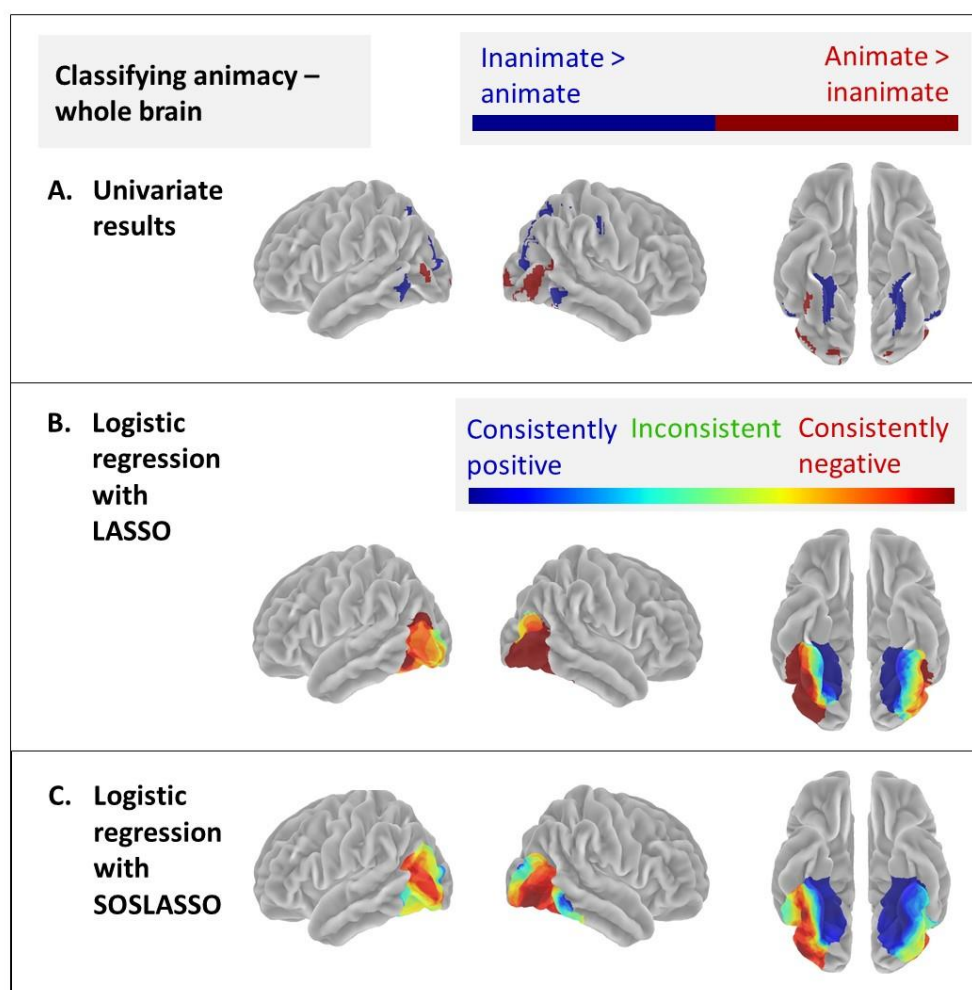


Figure 5: Coefficients for classification analyses. (A) Univariate effects ($A > I$, red; $I > A$, blue) are shown projected to the cortical surface and are cluster-corrected at $p < 0.05$ based on an uncorrected voxel threshold of $p < 0.001$. (B) Coefficients for logistic regression classifiers trained with LASSO regularisation on contrast beta values to discriminate animate from inanimate stimuli are shown projected to the cortical surface and thresholded via permutation testing to control the false discovery rate at $\alpha = 0.05$ (see Methods). Warm colours indicate that coefficients are mostly negative across participants (since animals were coded as 0 and inanimate objects as 1, a negative coefficient indicates that a positive beta value is associated with increased probability that the stimulus is animate); cool colours indicate that coefficients are mostly positive across participants (a positive beta value is associated with increased probability that the stimulus is inanimate); green shades indicate that the region was selected in multiple participants but that the sign of the coefficient was not consistent across participants. (C) Coefficients for logistic regression classifiers trained with SOSLASSO regularisation on contrast beta values to discriminate animate from inanimate stimuli are shown projected to the cortical surface and thresholded via permutation testing to control the false discovery rate at $\alpha = 0.05$ (see Methods). The colour scaling is the same as in (A).

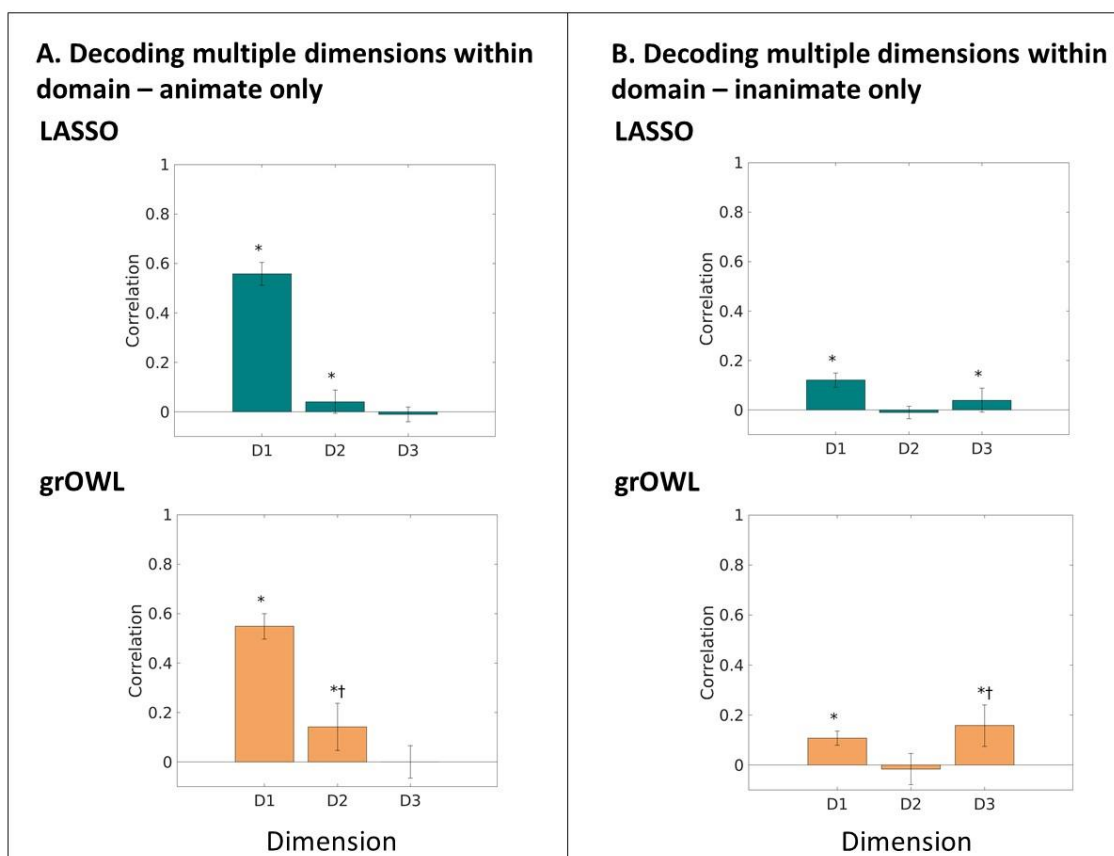


Figure 6: Hold-out correlations within-domain at the whole-brain level. Mean and 95% confidence interval of the hold-out correlation for RSL models trained on contrast beta values to predict the coordinates of held-out stimuli on three target semantic dimensions (dimension 1, D1; dimension 2, D2; and dimension 3, D3). Models are trained using LASSO regularisation (top, teal) or using grOWL regularisation (bottom, orange). Stars (*) indicate a significant correlation between predicted and target coordinates (as determined by permutation test, probabilities adjusted to control the false-discovery rate at $\alpha = 0.05$). Daggers (†) indicate a significant difference between correlation on that dimension using that regularisation penalty and correlation on the same dimension using the other regularisation penalty (probabilities adjusted to control the false-discovery rate at $\alpha = 0.05$). (A) Hold-out correlations for animate stimuli only; (B) Hold-out correlations for inanimate stimuli only.

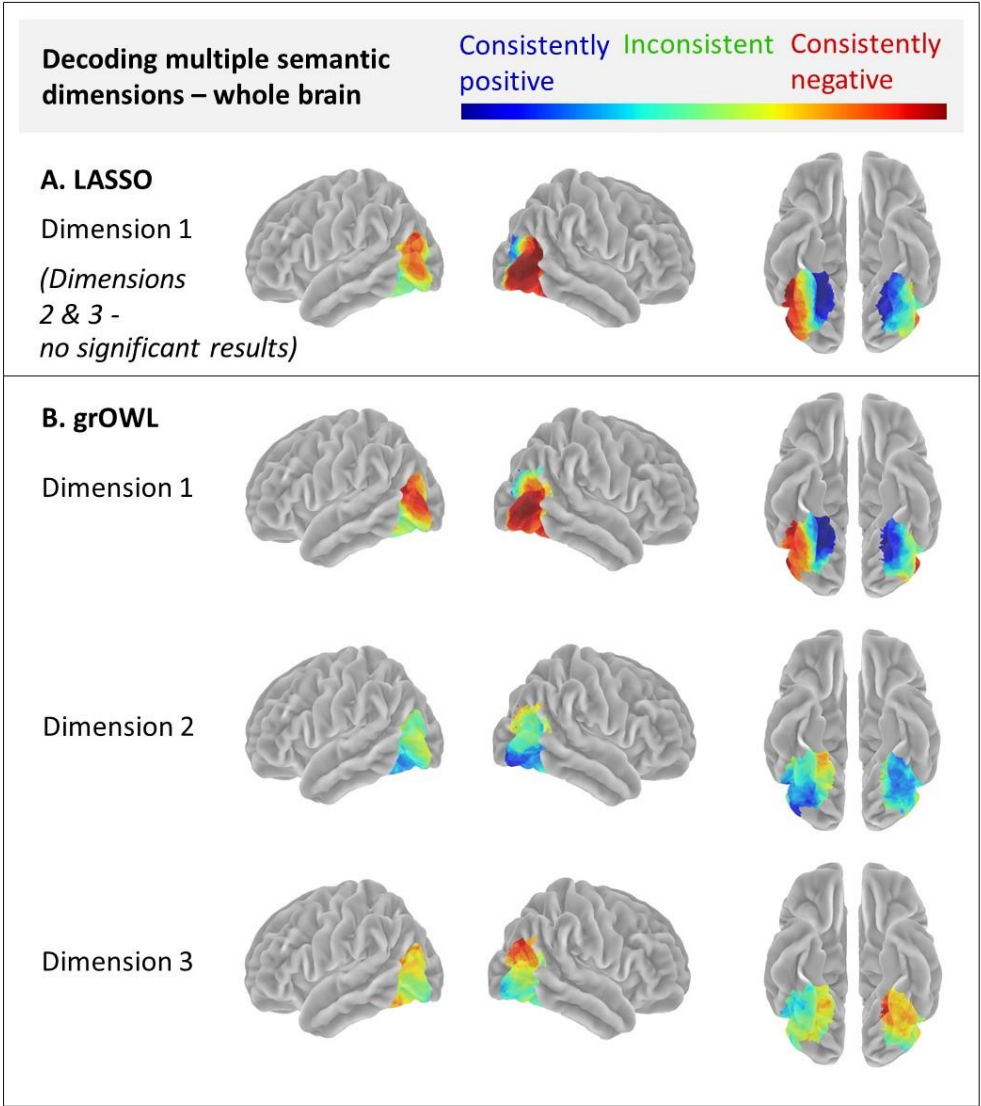


Figure 7: Coefficients for RSL analyses. (A) Coefficients for RSL models trained with LASSO regularisation on contrast beta values to predict the coordinates of held-out stimuli on three target semantic dimensions are shown projected to the cortical surface and thresholded via permutation testing to control the false discovery rate at $\alpha = 0.05$ (see Methods). Warm colours indicate that coefficients are mostly negative across participants; cool colours indicate that coefficients are mostly positive across participants; green shades indicate that the region was selected in multiple participants but that the sign of the coefficient was not consistent across participants. (B) Coefficients for RSL models trained with grOWL regularisation on contrast beta values to predict the coordinates of held-out stimuli on three target semantic dimensions are shown projected to the cortical surface and thresholded via permutation testing to control the false discovery rate at $\alpha = 0.05$ (see Methods). The colour scaling is the same as in (A).

3. Discussion

Equipped with a novel 7T-fMRI acquisition protocol capable of recording signal in ventral temporal regions²⁶ and four innovative decoding methods^{11,28–31}, this study detected, characterised, and thereby adjudicated theories about the brain’s semantic representations. We revealed graded, multidimensional semantic structure in the vATL, represented by an interdependent, contiguous, dynamic⁴⁴ code. We also discovered interdependent, heterogeneous, inconsistent²⁷ representations in posterior temporal and occipitotemporal cortex.

Multiple theories of semantic cognition implicate the vATL, but those theories differ in the neural code that they propose to exist in the vATL. Damasio and colleagues³ hold that the vATL contains convergence zones that simply coordinate representations elsewhere – the vATL does not itself express semantic information. Conversely, we could decode multiple semantic dimensions from vATL activity. Other theories argue that the vATL functions as a hub that assembles and encodes complex semantic representations. The finding that RSL with grOWL, which assumes that features are interdependent, outperformed LASSO indicates that the vATL represents semantic information as an uninterpretable vector-space code, aligning with predictions of the hub-and-spoke model^{6,9,41} rather than theories positing independent representational elements^{1,2,40}. Also in accordance with predictions of the hub-and-spoke model, poorer decoding performance was observed in the anterior than the posterior half of our vATL ROI, suggesting the existence of a dynamic code that is most clearly observed with time-resolved imaging methods.

Our ROI results converge with results from neuropsychology^{32–35}, TMS^{23,36–39}, intracranial electrophysiology^{28,44,59–64}, and computational modelling^{9,42,43}, yet diverge from other multivariate studies using fMRI, which rarely highlight the vATL^{10–18}. Two factors are likely to account for this difference. The first is signal inhomogeneity caused by the proximity of the vATL to the air-filled sinuses. Employing a novel multi-echo, multiband 7T-fMRI acquisition sequence²⁶, in contrast to the standard acquisition used by others, we were able to decode both animacy and graded, multidimensional semantic similarity structure even from the most anterior region of the vATL. Collecting data with a standard sequence (to enable a direct comparison) was beyond the scope of this study; previous work has

directly tested the link between acquisition sequence and correlation-based decoding accuracy²⁶ and others should extend this test to decoding methods that depend on machine learning (including logistic regression and RSL). The second factor is choice of method. This study illustrated clear differences between decoding methods – SOSLASSO produced higher classification accuracy than LASSO and grOWL decoded multiple semantic dimensions more reliably than LASSO. The regularisation penalties that we used make assumptions (sparsity and/or anatomical clustering and/or interdependence) that are explicitly based on theories about how neural systems represent information^{11,28}. However, note that, in comparison to common approaches such as encoding^{5,13,14} and RSA^{16–18}, the assumptions they make are few and gentle. Our results suggest that adopting mild, neurally-inspired assumptions may produce better coherence between fMRI results and results from other methods. However, as argued previously²⁷, the most appropriate method for future studies is the one that aligns with those individual researchers’ explicit hypotheses. In summary, aided by two technical advances, we have been able to reconcile the discrepancy between multivariate imaging and other work.

Our whole-brain analysis discovered graded, multidimensional semantic similarity structure in posterior temporal and occipitotemporal cortex. These regions have been implicated by multiple theories – some propose that they encode visual properties including colour, form, and motion^{7,8} that may be particularly relevant for representing tools⁴⁹; others even propose that they encode mental states⁵. Our results enable us to distinguish competing theories in two ways. First, as in the ROI analysis, we can adjudicate theories based on the *nature* of the code that they propose. Detecting codes of varying natures is an advantage of our multivariate approach²⁷ – note that the regions of inconsistent code direction (green in Figures 5B and 5C) were missing from the univariate analysis (Figure 5A). The interdependent, heterogeneous, inconsistent representations that we discovered match the *kind* of code predicted by the hub-and-spoke model^{6,9,41} (see also¹⁵). Note, however, that the hub-and-spoke model does not predict what informational content we should expect in the spokes (a set of regions which may include the posterior temporal and occipitotemporal cortex). Preliminary evidence from computational modelling suggests that spoke regions do come to represent overall semantic similarity structure (rather than similarity structure within a particular modality) during semantic tasks^{42,50}, but it may only be the spoke that is processing the current sensory input that does so⁶⁵. The results of this

study are consistent with these exploratory findings – we presented visual stimuli and found similarity structure in visual association cortex.

A second method of distinguishing between theories is considering which regions were *not* selected in our analysis. The angular gyrus, which has been proposed as an additional semantic hub^{1,2}, was absent, which means that our results do not readily support a multi-hub perspective. The differences between this study and previous work^{1,66,67}, however, remain to be explored in detail. One possible explanation is that the angular gyrus is not a semantic store, but instead buffers multimodal and/or temporally-extended information, including semantic information, during cognition^{54,68}. Alternatively or additionally, the angular gyrus may contribute to semantic control, possibly by representing a semantic space that is “warped” to emphasise task-relevant dimensions⁶⁹. To test this, future studies should employ paradigms that make greater control requirements than our naming paradigm does.

The vATL was also absent from our whole-brain results, despite the fact that our ROI analysis detected semantic structure there. This discrepancy may arise because the permutation distribution used for statistical thresholding of these maps is nonuniform – voxels in the vATL are selected far more frequently than voxels elsewhere (Supplementary Figures S1 and S2). This methodological quandary remains unresolved, but preliminary simulation results suggest that overselection occurs when the correlated noise structure present in fMRI breaks down under conditions of magnetic inhomogeneity, increasing the dimensionality of the data and, consequently, the probability that data will correlate with a permuted target (Frisby, unpublished analyses). Future work should both probe this problem further and evaluate alternative methods of statistical significance testing for decoding.

5. Conclusion

Previous theoretical work described the kinds of semantic representations that might possibly exist²⁷. This study represents the first endeavour to adjudicate those possibilities empirically. We revealed graded, multidimensional semantic structure in the vATL, represented by an interdependent, contiguous, dynamic⁴⁴ code. We also discovered interdependent, heterogeneous, inconsistent²⁷ representations in posterior temporal and

occipitotemporal cortex. These results align better with the hub-and-spoke model of semantic cognition than with competing theories. Our comparative multivariate approach is an effective method that has the potential to reveal how the brain represents, not only semantic knowledge, but any kind of information.

6. Online methods

6.1. Participants

32 participants were recruited to the study. Of these, five were unable to complete the study either for medical reasons (for example, an incidental finding) or because of technical problems during data acquisition which meant that the scanner needed to be rebooted. This left 27 participants (age range 19-50, mean age 27.96 years, 17 female, 10 male) who were all native speakers of British English, were right-handed, had normal or corrected-to-normal vision, and had no neurological or sensory disorders. All participants gave written informed consent and the research was approved by a local National Health Service (NHS) ethics committee.

6.2. Stimuli and task

Stimuli were the same 100 line drawings used in previous ECoG work^{28,44,59} – 50 animals and 50 inanimate items including vehicles, clothes, musical instruments and a range of other objects⁷⁰ (<https://github.com/slfrisby/7TConvergent/tree/main/stimuli/>). There were no significant differences between the categories with respect to age of acquisition, visual complexity, familiarity, word frequency, name agreement, and non-semantic visual structure^{44,71}. PsychoPy 2022.2.5⁷² was used to display stimuli and to record speech. Stimuli for both tasks were rear-projected onto a screen in the MRI scanner, and observers viewed stimuli through a mirror mounted to the head coil directly above the eyes.

The study had a fast event-related design with 4 runs per participant, collected in a single session. Each run began 16 s after the start of the MR sequence and consisted of 100 trials (each stimulus appeared once per run). Each trial consisted of a stimulus presented for 4 s followed by a fixation cross presented for 4 s. 4 sequences of animate and inanimate stimuli were created and counterbalanced across participants. The order of trials was optimised for the contrast between animate and inanimate stimuli using optseq2

(<https://surfer.nmr.mgh.harvard.edu/fswiki/optseq2>). For each run and each participant, a random order of animate stimuli and a random order of inanimate stimuli were created; the stimuli were then interspersed according to the sequence generated with optseq2.

Participants were instructed to name each picture out loud as quickly and accurately as possible while keeping head motion to a minimum. To ensure that participants understood the instructions, they practised the task before entering the scanner while the experimenter provided feedback about their head motion. Practice continued until the experimenter was satisfied that head motion was minimised.

Overt speech was recorded for each trial and responses were scored for accuracy with the study's aim of decoding semantic similarity structure in mind. If the participant did not respond within the 4 seconds during which the stimulus was on-screen, the trial was marked as incorrect. If the participant responded incorrectly but named an object that the rater judged to be visually and semantically similar to the stimulus and at a similar level of specificity (e.g. "wasp" for bee), the trial was marked as correct; if the object that they named was semantically dissimilar or was named at a less specific level (e.g. "insect" for bee), the trial was marked as incorrect. It was important that neural responses did not reflect a "blend" of the semantics of multiple stimuli so, if the participant gave two synonyms (e.g. "gun, pistol") the trial was marked as correct; if they named two different objects ("kettle, iron") the trial was marked as incorrect even if one of the names was correct. Trials containing fillers such as "um" and "err" in addition to a correct response were marked as correct because the participants were neurologically healthy and there was no reason to believe that these utterances reflected anything other than normal speech processes. The mean and standard deviation of the accuracy scores were calculated.

6.3. Image acquisition

All images were acquired on a whole-body 7T Siemens MAGNETOM Terra MRI scanner (Siemens Healthcare, Erlangen, Germany) with an 8Tx32Rx head coil (Nova Medical, USA).

An MP2RAGE anatomical scan was acquired with the following parameters: 224 sagittal slices (interleaved acquisition), FOV 240 x 225.12 x 240 mm, voxel size 0.75 x 0.75 x 0.75 mm, TR 4300 ms, inversion times 840 and 2370 ms, TE 1.99 ms, flip angles 5° and 6°, GRAPPA acceleration factor 3. Next, manual B₀-shimming was performed over the volume

to be used for echo-planar imaging (EPI). The aim was to reduce the water linewidth, defined as full width of the spectrum at half height, to below 40 Hz. However, for some participants the adjustments proved time-consuming and so adjustment time was capped at 30 minutes from the start of scanning after which the best shim parameters were adopted. The actual average water linewidth was 38.80 Hz (standard deviation = 5.93 Hz).

All four functional runs of EPI used the multi-echo multiband sequence optimised by Frisby and colleagues²⁶ with the following parameters: 48 axial slices (interleaved acquisition), FOV 210 x 210 x 210 mm to cover the whole brain in most participants (visual inspection ensured that the vATL was included in the FOV for all participants and the FOV was tilted up at the nose to avoid ghosting of the eyes into the vATL), TR = 1510 ms, 3 echoes, TE₁ = 11.80 ms, TE₂ = 27.05 ms, TE₃ = 42.30 ms, multiband factor 2, GRAPPA acceleration factor 3, phase partial Fourier 7/8, voxel size 2.5 mm isotropic (no gap), flip angle 63°, and A-P phase encoding direction. Immediately *before* each run, 5 additional volumes were acquired with the phase encoding direction changed to P-A. These volumes were to be used for distortion correction during preprocessing.

6.4. Data analysis

All analysis code is available at: <https://github.com/slfrisby/7TConvergent/>.

6.4.1. Preprocessing

For ease of data sharing we converted all DICOMs to BIDS format⁷³ using *heudiconv* (v1.1.1)⁷⁴.

Standard reproducible preprocessing pipelines designed for 3T-fMRI, such as *fMRIPrep*^{75–77}, perform poorly on 7T-fMRI EPI data; therefore, the analysis pipeline was split into stages using different software packages. The MP2RAGE T1w (combined) image first had its background noise removed using O’Brien regularisation⁷⁸ and was then submitted to CAT12 for segmentation⁷⁹ (in SPM12; <https://www.fil.ion.ucl.ac.uk/spm/>). The bias-corrected and global-intensity-corrected T1w (combined) image produced was provided to as input to *fMRIPrep* 21.0.1. The T1w (combined) image was corrected for intensity non-uniformity with *N4BiasFieldCorrection*⁸⁰ and skull-stripped with a Nipype implementation of

the *antsBrainExtraction.sh* workflow (ANTs). Brain tissue segmentation of cerebrospinal fluid (CSF), white matter (WM), and grey matter (GM) was performed on the brain-extracted T1w (combined) image using *fast* (FSL 6.0.5.1)⁸¹. Volume-based spatial normalisation to standard space (MNI152NLin2009cAsym) was performed through nonlinear registration with *antsRegistration.sh* (ANTs 2.3.3, <https://github.com/ANTsX/ANTs>)^{82,83}.

Functional preprocessing was performed using in-house code, composed of functions from AFNI (v.18.3.03; <https://afni.nimh.nih.gov/>)^{84,85}, FSL (v.5.0.10; <https://fsl.fmrib.ox.ac.uk/fsl/fslwiki>)^{86–89}, tedana (v.24.0.0; <https://tedana.readthedocs.io/en/stable/index.html>)^{90–93}, and ANTS (v. 2.2.0; <https://github.com/ANTsX/ANTs>)^{82,83}. EPIs were slice-time corrected to the middle slice using *3dTshift* (AFNI). Next, EPIs from the first echo were motion corrected with *3dvolreg* and *3dAllineate* using the first volume of the first run acquired as a reference. The same transforms were applied to the EPIs from the second and third echoes. EPIs were then skull-stripped using *BET* (FSL)⁸⁸ to create a participant-specific brain mask. *tedana* was used to combine data from all echoes optimally based on T2* weighting⁹⁴ and to denoise the data using ICA. *tedana* conducts denoising by decomposing data using PCA and ICA, classifying components according to whether the signal scales linearly with TE (as BOLD signal does), and reconstructing the data using only BOLD-like components. The brain mask created with *BET* was used as the mask for this stage.

Next, unwarping was conducted using *topup* and *applytopup* (FSL)⁹⁵. Field displacement maps were calculated using 10 volumes (5 with A-P phase encoding direction, extracted from the start of each functional run, and 5 with P-A phase encoding direction, collected separately before each run) and the resulting correction was applied to all images.

Finally, the mean EPI for each run was coregistered to the skull-stripped native structural image using a rigid-body registration in *AntsRegistrationSyN.sh* (ANTs). This produced transforms from native EPI space to native T1 space. For all decoding analyses, images remained in native space and were not smoothed. However, the transforms from native EPI to native T1 (for the first run) and the transforms from native T1 to standard space (MNI152NLin2009cAsym), calculated with *antsApplyTransforms* (ANTs), were used to calculate the MNI-space coordinates that each voxel *would* have *were* images to be transformed into MNI space. (These coordinates were used for visualisation and were provided to classifiers trained with SOSLASSO regularisation, which take anatomical

information as input.) The transform from native EPI to native T1 space was also used to transform the participant's grey-matter mask into native EPI space. For the univariate analysis, images were transformed into standard space (MNI152Nlin2009cAsym) with *antsApplyTransforms* (ANTs) and smoothed with a 6 mm FWHM Gaussian filter in SPM12 (<https://www.fil.ion.ucl.ac.uk/spm/>).

6.4.2. 1st-level (within-participant) GLM

Data were processed using the general linear model (GLM) approach implemented in SPM12 in MATLAB r2023b. For decoding, each stimulus was modelled as an individual boxcar function, 1.5 seconds in length because the stimulus in our stimulus set with the longest mean naming time had a mean naming time of 1323 ms⁷⁰ (periods between stimuli were modelled implicitly). These boxcar functions were subsequently convolved with SPM's difference of gammas haemodynamic response function. The six motion parameters extracted during preprocessing were used as regressors of no interest. Data from all four runs were concatenated, so an additional constant was included for each run. The micro-time resolution was set as the number of slices ($n = 48$), the micro-time onset was set as the reference slice for slice-time correction ($n = 24$), and the high-pass filter cutoff was 128 seconds. Each participant's grey matter mask in native EPI space was used as a mask for the analysis. The parameter estimation method was restricted maximum likelihood estimation (ReML) and serial correlations were accounted for using an autoregressive AR(1) model during estimation. This produced one beta image per stimulus in native space. For univariate analysis, the same model was run on the normalised, smoothed data with identical parameters except that a grey matter mask in standard space (MNI152Nlin2009cAsym) was used and that stimuli named incorrectly were modelled as additional regressors of no interest. The univariate contrasts of interest were greater activation for animate stimuli named correctly than inanimate stimuli named correctly ($A > I$) and greater activation for inanimate stimuli named correctly than animate stimuli named correctly ($I > A$). This produced contrast images in MNI space.

6.4.3. Univariate analysis – 2nd-level (across-participant) GLM

Group-level t-tests of each contrast of interest ($A > I$, $I > A$) were conducted in SPM12. The group t-maps were assessed for significance by using a voxel-height threshold of $p < 0.001$ to define clusters and then a cluster-defining family-wise-error-corrected threshold of $p < 0.05$ for statistical inference. Group-level results were then projected to the pial surface (fsaverage template^{96,97}) using *-volume-to-surface-mapping* with trilinear interpolation in Connectome Workbench 1.5.0 and the thresholded maps were plotted on the pial surface using *plot_surf_stat_map* in *nilearn* implemented in Python 3.9.

6.4.4. Multivariate decoding

To prepare data for decoding, beta images from all 4 presentations of each stimulus were averaged across runs. If a participant had named a stimulus incorrectly on one to three occasions, the beta image(s) corresponding to the incorrect trial(s) were not included in the averaging. If the participant named the stimulus incorrectly on all four trials, the values for each voxel were interpolated with the median value for that voxel over all other stimuli.

We constructed our vATL ROI based on electrode coverage in a previous study⁵⁹ (all electrode coordinates are available at <https://github.com/slfrisby/7TConvergent/>). Electrode coordinates in the temporal lobe were converted to a single NIFTI volume at native resolution (2.5 mm isotropic) in MNI space using SPM12 and smoothed at 10mm FWHM. In order to exclude regions that lay outside the temporal lobe, a mask was created by taking the Harvard-Oxford cortical and subcortical atlases, removing the temporal lobe from the atlas, and adding the cerebellum from the AAL 2 atlas⁹⁸ (since the Harvard-Oxford atlas does not include a cerebellum). Only voxels that lay outside this atlas mask were included in the ROI. After masking the ROI was smoothed again at 4mm FWHM because, on inspection, this created a smooth shape with space around the temporal lobes – since the ROIs were to be backprojected into native space, it was important to ensure that individual participants' temporal lobes would be fully encompassed. Since electrode coverage was far more extensive in the left than the right hemisphere, a mirror-image of the left ROI was used as the right ROI. To create half-ROIs to enable tests for dynamism, we bisected the ROIs between their anteriormost and posteriormost points ($y = -20$). The final ROIs in standard space (MNI152NLin2009cAsym) are shown projected to the surface in Figures 1 and 3 and are available at <https://github.com/slfrisby/7TConvergent/tree/main/ROI/>.

The ROIs were backprojected into each participant’s native space using the transforms from native EPI to native T1 and the transforms from native T1 to standard space generated with *antsRegistration.sh*. For the ROI decoding analysis, only beta values from voxels that overlapped with both the ROI and the participant’s native-space grey matter mask were provided as input to decoding models. (For the whole-brain analysis, only voxels within the participant’s native-space grey matter mask were provided as input).

We tested for semantic structure using two decoding methods.

6.4.4.1. Regularised logistic regression

We used regularised logistic regression to decode animacy. Regularised logistic regression classifiers avoid overfitting by finding a set of coefficients β (one per feature, in this case per voxel) that minimise both prediction error and a regularisation penalty:

$$\min_{\beta} ((1 - \lambda)f(\beta) + \lambda h(\beta))$$

where $f(\beta)$ is the prediction error, $h(\beta)$ is the regularisation penalty, and λ is a learnt hyperparameter that controls how the optimisation should weigh the regularisation penalty versus the prediction error.

One set of classifiers was trained using LASSO (L1) regularisation⁵⁵, for which the regularisation penalty is:

$$h(\beta) = \sum_{j=1}^m |\beta_j|$$

The LASSO penalty increases as the sum of the absolute values of the coefficients increases and is therefore at its lowest when most coefficients are zero and the voxels that receive the few nonzero coefficients are uncorrelated in their activity. Classifiers with LASSO therefore encapsulate the assumption that the neural code is sparse (only a few voxels encode the target structure) and uncorrelated.

Classifiers with LASSO were fitted using *glmnet* in MATLAB r2023b⁹⁹, using 3 – 0.2 as the range of possible lambda values. Each classifier took, as input, beta values from multiple voxels from a single participant.

A second set of classifiers was trained using SOSLASSO regularisation^{11,30,31}. Before model training, voxels are assigned to “sets” – ROI-like spheres defined in MNI space. In this

study, the sets were 9 mm in radius and overlapped by 9mm with adjacent spheres. All decoding analyses were conducted in native space, but each voxel was assigned to sets based on the location that it *would* have *were* the images to be transformed into MNI space (see Preprocessing). The sets are defined in a common space, so voxels from multiple participants are assigned to the same anatomical set.

Classifiers are then fit to beta values from multiple voxels from *all* participants. The SOSLASSO regularisation penalty is calculated for each set individually and summed over all sets S :

$$h(S, \beta) = \sum_S \left((1 - \alpha) \sum_{j=1}^m |\beta_j| + \alpha \sqrt{\sum_{j=1}^m \beta^2} \right)$$

where α is a learnt hyperparameter that scales the two components of the SOSLASSO penalty – the LASSO penalty $\sum_{j=1}^m |\beta_j|$, which increases as the sum of the coefficients increases, and the grouping penalty $\sqrt{\sum_{j=1}^m \beta^2}$, which also increases as the sum of the coefficients increases but increases more steeply when those voxels are in different sets than when they are in the same set. This means that classifiers with SOSLASSO assume that the code is sparse, but also found in similar locations within and across participants.

Classifiers with SOSLASSO were fitted using the WISC MVPA toolbox in MATLAB r2018b (https://github.com/crcox/WISC_MVPA/) with hyperband budget set to 25 (all other parameters were set as defaults, including 3 – 0.2 as the range of possible lambda values).

6.4.4.2. Representational similarity learning

Next, we used RSL to decode graded, multidimensional similarity structure. We first constructed a representational similarity matrix (RSM) S that expressed the semantic similarity between each possible pair of stimuli (operationalised using feature verification norms⁵⁶). We then applied singular value decomposition to extract three components accounting for 89.5% of the variance in the whole RSM (81.1%, 4.4%, and 4.0%, respectively)²⁸. Figure 2 shows the coordinates of all 100 stimuli on each dimension. Where n is the number of stimuli and r is the number of dimensions, SVD yields three products:

1. An $n \times r$ matrix ($\mathbf{U}$) of components, also known as singular vectors.
2. The transpose of $\mathbf{U}$ ($\mathbf{U}^T$).
3. An $r \times r$ matrix ($\mathbf{D}$) of singular values.

To summarise:

$$\mathbf{S} \cong \mathbf{U}\mathbf{D}\mathbf{U}^T$$

We weighted the components $\mathbf{U}$ by the square root of the singular values $\sqrt{\mathbf{D}}$ to create an $n \times r$ matrix $\mathbf{C}$ – three target dimensions for decoding.

$$\mathbf{C} = \mathbf{U}\sqrt{\mathbf{D}}$$

Next, RSL models were trained to predict the coordinates of each stimulus on each dimension. Each voxel receives three coefficients – one per dimension – and so the set of coefficients to be learnt is an $m \times r$ matrix $\boldsymbol{\beta}$. Where $\mathbf{X}$ is the matrix of data (of size n stimuli $\times$ m voxels) and α is a learnt hyperparameter, RSL aims to find the set of coefficients that minimises:

$$\min_{\boldsymbol{\beta}} ((1 - \lambda) \|\mathbf{C} - \mathbf{X}\boldsymbol{\beta}\|_F + \lambda H(\boldsymbol{\beta}))$$

One set of models used LASSO regularisation and were trained using *glmnet* in MATLAB r2023b, using 6 – 0 as the range of possible lambda values. Each model took, as input, beta values from multiple voxels from a single participant.

A second set of models used grOWL regularisation. grOWL works by creating a vector w of non-negative, non-increasing values (i.e. each successive value of w is smaller than or equal to the previous value). w is of length m (the number of voxels) and a correspondence is learnt between elements of w and rows of $\boldsymbol{\beta}$ such that the row of $\boldsymbol{\beta}$ containing the largest coefficients (defined by the Euclidean norm or 2-norm of those coefficients) receives the largest value in w , the row containing the next largest coefficients receives the next largest value, and so on. Note that, because successive elements of w can be equal, rows of $\boldsymbol{\beta}$ containing very similar coefficients can receive the same value in w .

The total grOWL regularisation penalty is calculated by taking the Euclidean norm (also known as the 2-norm) of each row of $\boldsymbol{\beta}$, multiplying that with its corresponding value in w , and summing the results. Where $\boldsymbol{\beta}_{[i]}$ is the row of $\boldsymbol{\beta}$ with the i -th largest 2-norm and w_i is the value in w corresponding to $\boldsymbol{\beta}_{[i]}$:

$$H(\beta) = \sum_{i=1}^m w_i \|\beta_{[i]}\|_2$$

$\|\beta_{[i]}\|_2$ increases as each coefficient increases. However, unlike LASSO which prefers voxels to be uncorrelated in their activity, grOWL assigns similar values in w to correlated voxels, meaning that those correlated voxels either all receive nonzero coefficients or all receive coefficients of zero. Consequently, grOWL assumes that the neural code is sparse but correlated – in other words, while LASSO assumes that only a few voxels will be important, grOWL assumes that only a few *groups of correlated voxels* will be important. Note also that $\|\beta_{[i]}\|_2$ is smaller when coefficients in a row of β are similar in size than when one is large and others are small or zero. grOWL therefore makes the *spanning assumption* – that important voxels should predict multiple target dimensions at once, and therefore that the target dimensions are represented interdependently.

grOWL models were trained using the WISC MVPA toolbox in MATLAB r2018a with hyperband budget set to 25 (all other parameters were set as defaults, including 6 – 0 as the range of possible lambda values). Each model took, as input, beta values from multiple voxels from a single participant.

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6.4.4.3. Cross-validation and model assessment

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Classifier or model accuracy was assessed via ten-fold nested cross-validation. In each outer loop, ten out of 100 stimuli (five animate, five inanimate) were defined as the outer-loop hold-out set. A second ten stimuli (five animate, five inanimate) were defined as the inner-loop hold-out set. The remaining 80 stimuli were used as training data to search for the best hyperparameter(s) - λ , plus α for SOSLASSO and w for grOWL. Hyperparameter performance was assessed on the inner-loop hold-out set. The folds were then reassigned so that ten stimuli previously used for training became the inner-loop hold-out set. Once all combinations of inner-loop hold-out set and training data had been explored, the best-performing hyperparameter value(s) were used to train a final model that was assessed on the outer-loop hold-out set. The procedure was then repeated with the nine other possible hold-out sets. In the first instance, stimuli were assigned randomly to the ten hold-out folds,

but this order was then fixed so that models fitted with every kind of regularisation were assessed using the same fold configuration.

The mean classification accuracy (for regularised logistic regression classifiers) or correlation between true and predicted coordinates (for RSL models) was calculated over all hold-out folds. Statistical significance was assessed via permutation testing. This was done for two reasons. First, since we used a fast event-related design, the haemodynamic response overlapped between trials. This means that, if temporally adjacent stimuli were divided between training and test sets, the stimulus in the training set may contain information about the stimulus in the test set and so may produce classification hold-out accuracies above 0.5 or hold-out correlations above 0. Second, cross-validated correlation (as was calculated for RSL models) is known to produce predicted coordinates that sometimes correlate *negatively* with the target coordinates in a hold-out set – the mean correlation under the null hypothesis may be below zero^{28,100}; therefore, the magnitude of the real correlation can be understood only with reference to the permutation distribution. In permutation testing, the stimulus labels were permuted before model training (as for hold-out folds, the permutation orders were generated at random but then fixed so that the permutations for each model were comparable). The procedure was repeated 100 times to produce 100 simulated accuracy or correlation values for each participant. A group-level empirical null distribution was simulated by randomly sampling one of the 100 values from each participant and computing a group average 10,000 times¹⁰¹. One-tailed p-values were calculated (where m is the number of values in the permutation distribution and b is the number of values in the permutation distribution larger than the true value^{28,102}):

$$p = \frac{b + 1}{m + 1}$$

To test whether one model or ROI produced a higher accuracy or correlation than another, a group-level permutation distribution of differences was created using the differences between the values in both permutation distributions and using the group-level simulation procedure described above. A p-value was then calculated for the true difference.

Statistical significance for all tests was defined after adjusting p-values to control the false discovery rate at $\alpha = 0.05$ ¹⁰³.

Ten-fold nested cross-validation does not provide a clear picture of which voxels are informative – each of the ten final models selects its own set of coefficients¹¹. Therefore, to visualise coefficients, one model was trained for each participant using an inner loop of ten stimuli to identify the best hyperparameter(s) but no outer loop (i.e. all 90 stimuli that were not in the current inner loop hold-out set were used for hyperparameter exploration; therefore, all 100 stimuli were used for training). 100 permutation models were also trained for each participant. Coefficients were converted to one NIFTI volume per participant at native resolution (2.5 mm isotropic) in MNI space using SPM12. Coefficients both for models trained on real data and for models trained on permuted data were projected to the pial surface (fsaverage template^{96,97}) using *-volume-to-surface-mapping* with trilinear interpolation and smoothed on the surface at 6 mm FWHM using *-metric-smoothing* in Connectome Workbench 1.5.0. Two metrics were calculated – (1) the proportion of participants in which each vertex was selected and (2) the proportion of negative coefficients that each vertex received (since animals were coded as 0 and inanimate objects as 1, a negative coefficient indicates that a positive beta value is associated with increased probability that the stimulus is animate). The map of the proportion of negative coefficients was thresholded using the map of coefficient selection - vertices were included in the map only if they were selected significantly more frequently in the real distribution than they were in the permutation distribution, as assessed via binomial tests with p-values adjusted to control the false discovery rate at $\alpha = 0.05^{103}$. Finally, the thresholded map was plotted on the pial surface using *plot_surf_stat_map* in *nilearn* implemented in Python 3.9.

Data and code availability: Data will be made publicly available upon peer review and acceptance. Code is publicly available at: <https://github.com/slfrisby/7TConvergent/>.

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